Ahmedabad University

Brain MRI Tumor Segmentation and Quantitative Analysis Using Classical Image Processing

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The Medics

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Motivation Behind the Project

- **Why This Problem:**
  - Brain Tumors are one of the most deadliest forms of ailments a person can go through.
  - An early screening and detection can help a person be safe from danger.
  - With Time, there has also been a rise in cases related to brain tumors.
  - We want to help the society in a form that would be massively helpful for a lot of the populace.
  - We aim to do good towards our fellow people through this idea, so that people do not have to see their friends and family in a hurtful state.

Problem Definition

Using Classical Image Processing Methods like Filtering and Segmentation to detect Brain Tumors in MRI Scans.

Literature Review

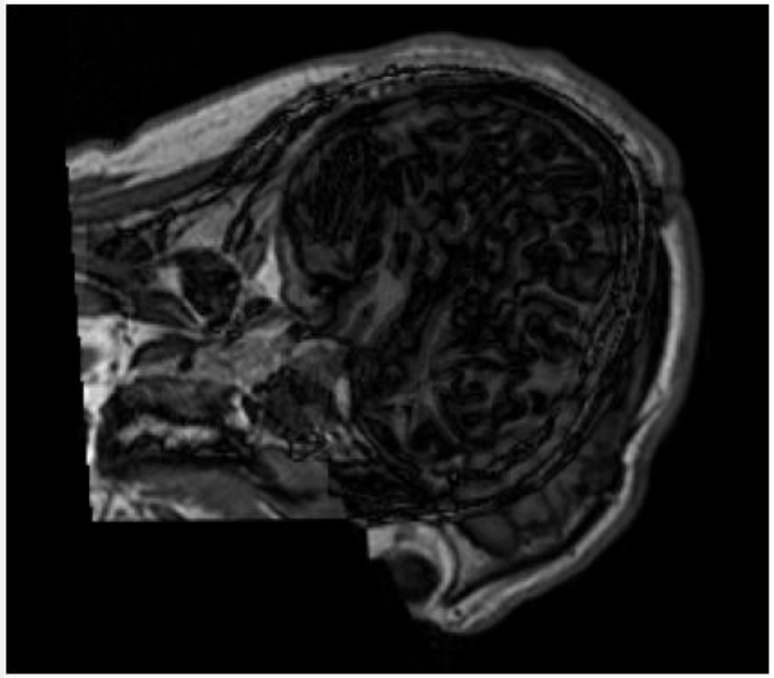
- **Magnetic resonance image-based brain tumour segmentation methods: A systematic review:**
  - This paper helped us in understanding how the entire system worked, This itself was a review paper so it helped us look into various methods we could go for.
  - However those methods were more so in the domain of ML & DL, then classical methods.
- **Rethinking Brain Tumor Segmentation From the Frequency Domain Perspective:**
  - We thought that they would have converted the domain to frequency like we had learnt in class.
  - They were also using Deep Neural Networks, so we at that point realised that we could not try to recreate any of those methods anyways.
- **BRISC: Annotated dataset for brain tumor segmentation and classification:**
  - This Paper we read parallely was more so to acquire another dataset to work on, they had collected many samples of MRI scans too, which we felt would be helpful to run our own code and inference on so that we could better understand our own progress on the same.

Methodology

- Direct implementation of basic filters like Laplacian
- Use of whole scan as a kernel
- Python overheads are one issue
- Use of CUDA with C/C++ for computation as the dataset is huge

Results

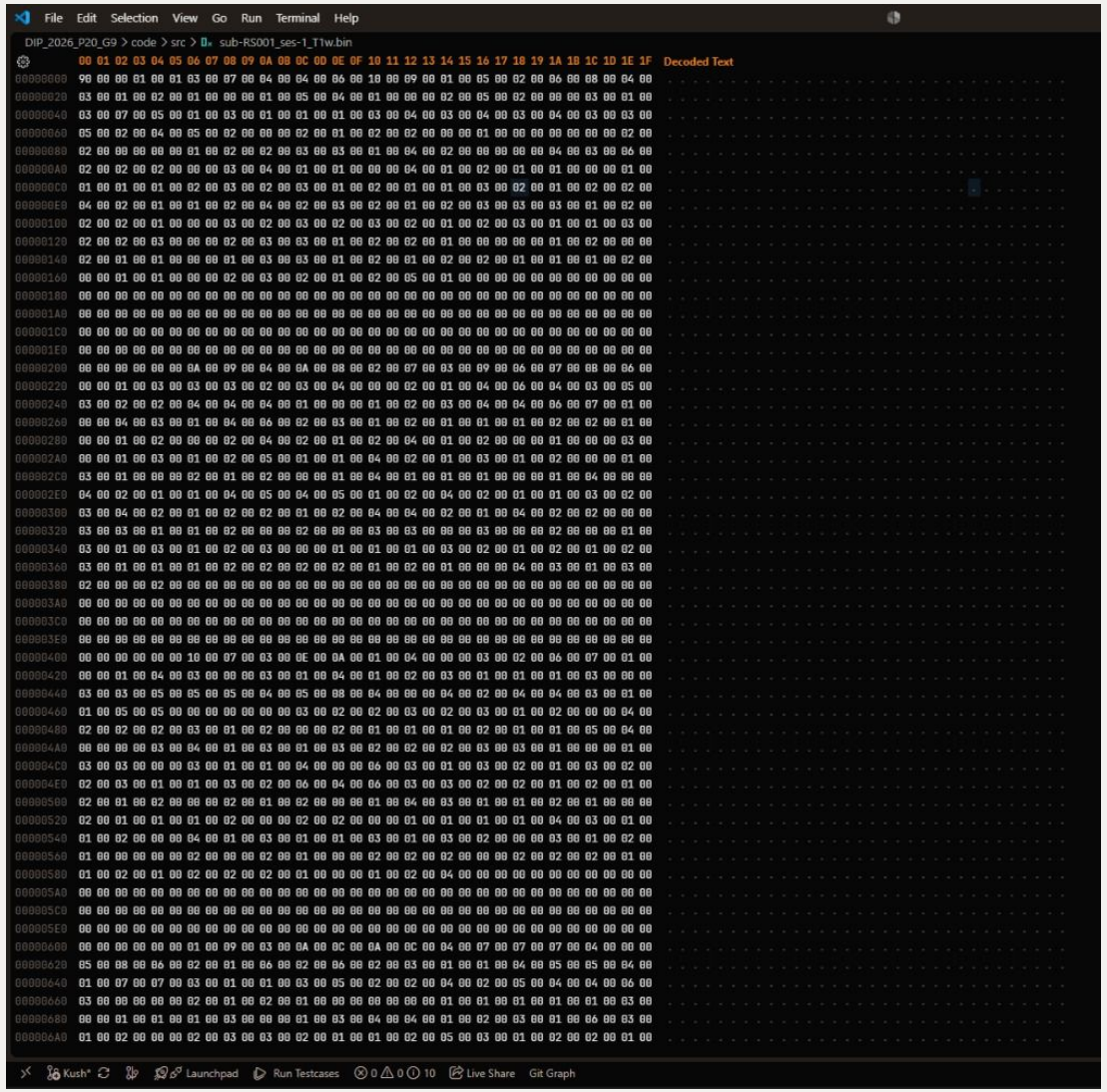




- This above image is result of base filtering which we had applied on the MRI scans.
- Tumor is basically an edge difference anomaly in the brain.

- We had tried to take a whole healthy MRI scan as a kernel and try to superimpose it on the scans.
- In this case, the only challenge is to map two scans on each other with precision.

Results



- This image contains the RAW binary voxel data of .nii.gz file.
- Since we want to ensure that the results are correct, we are performing multiple rounds of baseline and multiple rounds of testing.
- For efficient computation, we are implementing C++ alongside Python.
- Python helps us open the MRI file and perform basic scaling and alignment.
- However, due to the computational limits of python and the limits of python to process multiple rounds fast and efficiently. Our main part of processing is written in C/C++ with the CUDA libraries.
- Final image rendering will be done in Python.

What We plan to do Next

- **Our Progress and Workplan:**
  - **Data Conversion:** Converting Python-generated data into binary format for C++ processing.
  - **Tumor Segmentation:** Targeting completion within the next two weeks.
  - **Image Restoration:** Planned after segmentation to improve scan quality and recovery.
  - **Clinical Understanding:** Helping patients understand tumor growth and supporting doctors in explaining findings.
  - **Interactive UI**: Developing a simple, user-friendly interface for exploring scans.

Contribution (Commitments)

Member	Method
Jash Odedara	Code Verification and debugging, Literature Review, Methodology, Communication & Conceptualisation and Documentation
Kush Kelaiya	Majority of the Code Writing, Literature Review, Methodology and some parts of the Documentation
Kritee Shah	Some of the Code Writing, Literature Review, Methodology, Literature Review and some of the Documentation
Rushin Modi	Some of the Code Writing, Literature Review, Discovery of Datasets and Methodology, Some of the Documentation

References

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M. Shao et al., ”Rethinking Brain Tumor Segmentation From the Frequency Domain Perspective,” in IEEE Transactions on Medical Imaging, vol. 44, no. 11, pp. 4536-4553, Nov. 2025, doi: 10.1109/TMI.2025.3579213.